

NinjaTrader Supply/Demand Indicator Upgrade – Absorption Detection

This document outlines the required modifications to integrate absorption detection with buy/sell signals into the existing SupplyDemandZones indicator.

1. Add Variables (Top of Script)

```
// ABSORPTION SETTINGS
private double volumeThresholdMultiplier = 1.5;
private int volumeLookback = 20;
```

2. Add Volume Calculation (Inside OnBarUpdate)

```
double avgVolume = SMA(Volume, volumeLookback)[0];
bool highVolume = Volume[0] > avgVolume * volumeThresholdMultiplier;
```

3. Add Absorption Logic (Inside Zone Loop)

```
bool inZone = High[0] >= z.Bottom && Low[0] <= z.Top;

if (inZone && highVolume && !z.Broken)
{
    double body = Math.Abs(Close[0] - Open[0]);
    double upperWick = High[0] - Math.Max(Open[0], Close[0]);
    double lowerWick = Math.Min(Open[0], Close[0]) - Low[0];

    // DEMAND ZONE → BUY
    if (z.Type == "Demand")
    {
        if (Close[0] > Open[0] && lowerWick > body)
        {
            Draw.ArrowUp(this, "BUY_" + CurrentBar, false, 0, Low[0] - TickSize * 2, Brushes.Lime);
        }
    }

    // SUPPLY ZONE → SELL
    if (z.Type == "Supply")
    {
        if (Close[0] < Open[0] && upperWick > body)
        {
            Draw.ArrowDown(this, "SELL_" + CurrentBar, false, 0, High[0] + TickSize * 2, Brushes.Red);
        }
    }
}
```

4. Notes for Coder

- Ensure logic is placed inside existing zone loop.
- Do not duplicate drawing tags incorrectly.
- Test on 1-minute GC chart.
- Adjust volumeThresholdMultiplier if needed.